

Elcan İsmayılov

Full Stack Developer

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Summary

Full-Stack Engineer specializing in real-time systems and hardware-software integration. Experienced in digital signal processing, building complex architectures, and implementing modern user interfaces. Achieved successful results in designing end-to-end data pipelines and monitoring platforms.

Education

- **Azerbaijan National Aviation Academy – Bachelor's Degree** — Azerbaijan, Baku
  Major: Telecommunication Engineering — 2020 – 2024
- **Azerbaijan Technical University – Master's Degree** — Azerbaijan, Baku
  Major: Information Security of Telecommunication Systems — 2024 – 2026
- **Code Space – Bootcamp** — Azerbaijan, Baku
  Major: Java Foundation — 2023 – 2024
- **Div Academy – Bootcamp** — Azerbaijan, Baku
  Major: Artificial Intelligence Engineering — 2025 – 2026

Work Experience

- **AzTV** – Senior Engineer **– October 2024 – Present** — **Azerbaijan, Baku**

Ensuring uninterrupted operation of radio broadcasting infrastructure and data integrity at the Central Apparatus complex, managing the satellite communication network, supervising server and network infrastructure, and ensuring high availability of broadcast networks.
In the Studio Support unit, ensuring uninterrupted transmission of audio and video signals for television broadcasting, monitoring broadcast equipment, and promptly resolving technical faults.

- **AzTV** – Presenter / Reporter (Intern) **– September 2025 – February 2026** — **Azerbaijan, Baku**

Preparing on-the-spot reports in a broadcast environment and providing technical and creative support for journalistic activities.

- **Silkway** – Communications Engineer (Intern) **– May 2023 – September 2024** — **Azerbaijan, Baku**

Conducting technical inspections and monitoring of onboard radar systems and radio-frequency equipment.

Technical Skills

Python, JavaScript/JSX, SQL, Scikit-learn, Pandas, NumPy, NLP, FFmpeg, Google GenAI, FastAPI, PostgreSQL, SQLAlchemy, WebSockets, REST API, WebAuthn, JWT, QR Integration, RBAC, React 19, Vite, Cloudflare, WAF Integration, GCP, AWS, Docker, Shell automation.

Selected Projects

CGrep

- Built and deployed an RF monitoring platform covering the full spectrum of Azerbaijan's geography in the 88–108 MHz FM band using RTL-SDR-type hardware.
- 7 regional SDRs process ~700 measurements per second, ~2.5 million records per day. Query time was reduced to 75ms using Recursive CTE / Loose Index Scan to optimize the data flow for the dashboards in use.
- Built a WebSocket-based chat, user presence, file sharing, and local server-side AI integration for natural-language queries over RF data.
- Security ensured with multi-factor authentication, QR login, role-based access control, and audit logs.

Cclip

- Developed a comprehensive AI-based video automation system for creating videos from source materials.
- Integrated speech transcription, AI-based clip generation and voiceover, subtitle creation, and rendering processes into a single infrastructure.

Languages

- English (B2-C1) | Russian (A1) | Azerbaijani (Native)